

Title : Colloborative Code Editor

Abstract

The **Collaborative Code Editor (CCE)** is a web-based platform designed to enable real-time collaborative coding, allowing multiple users to write, edit, and execute code in a shared environment. The platform supports multiple programming languages, including Python, JavaScript, Java, and C, and offers features such as user authentication, room management, real-time chat, and code synchronization. Users can join or create rooms with unique identifiers, collaborate on coding tasks, and submit code for execution via the **Judge0 API**, receiving immediate feedback on execution results.

The **frontend** of the application is developed using **React** with **TypeScript**, providing a dynamic and responsive user interface for seamless real-time interactions. The build tool **Vite** is utilized to ensure fast development and hot-reloading during coding sessions.

The **backend** is implemented using **Express**, responsible for handling user authentication, room creation, and WebSocket communication. The backend integrates with **MongoDB** for data storage and leverages **Redis** for task queue management and Pub/Sub communication, ensuring efficient real-time updates across all connected users.

Collectively, the frontend and backend technologies deliver a robust and scalable platform ideal for collaborative coding in educational, competitive, and software development environments.